

Chapter 2. PERMUTATIONS

2.2 Cycles



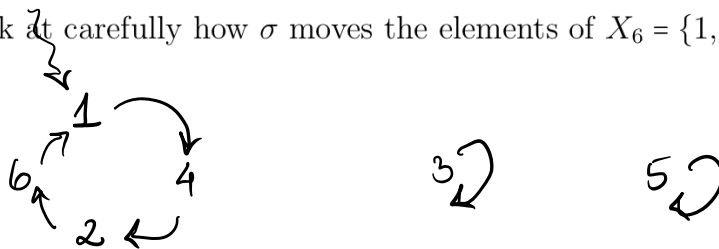
Today we will talk about a special type of permutations, called **CYCLES**. We will see that every permutation can be reconstructed using cycles.

How does a cycle look like?

Let us begin with an example: consider the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 6 & 3 & 2 & 5 & 1 \end{pmatrix} \in S_6$$

and look at carefully how σ moves the elements of $X_6 = \{1, 2, 3, 4, 5, 6\}$:



In other words, σ fixes 3 and 5, and moves 1, 2, 4, 6 in a cyclic way. Due to this reason, we call σ a **cycle**, and write it in a more compact way called the **cycle notation**: $\sigma = (1 \ 4 \ 2 \ 6)$. Notice that in the cycle notation

- we have only listed the elements that are moved by sigma;
- each element is moved to its right neighbor, except the last element, which “cycles around” to the first.

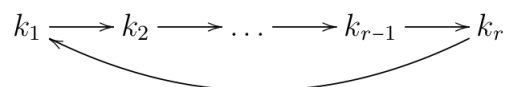
Definition 2.11. Let $k_1, k_2, \dots, k_r$ be distinct elements of X_n . A permutation $\sigma \in S_n$ fixing the remaining elements of X_n (if any) and such that

$$\sigma(k_1) = k_2, \quad \sigma(k_2) = k_3, \quad \dots, \quad \sigma(k_{r-1}) = k_r, \quad \sigma(k_r) = k_1,$$

called an **r -cycle**; r is called the **length** r of σ . We express σ as

$$\sigma = (k_1 \ k_2 \ \dots \ k_r),$$

which means that



$$\sigma = (4 \ 2 \ 6 \ 1) = (2 \ 6 \ 1 \ 4)$$

REMARKS

$$(k_1 \ k_2 \ \dots \ k_r) = (k_2 \ k_3 \ \dots \ k_r \ k_1) = \dots = (k_r \ k_1 \ k_2 \ \dots \ k_{r-1})$$

the **only** cycle of length 1 is the identity permutation,

a 2-cycle $(k_1 \ k_2)$ interchanges k_1 and k_2 , and fixes everything else; 2-cycles are called **transpositions**.

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 4 & 5 \end{pmatrix} = (1 \ 3) = (3 \ 1)$$

Examples 2.14. Write the following cycles into a cycle notation and indicate their length.

$$1. \ \sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 3 & 1 & 5 & 2 \end{pmatrix} \in S_5$$

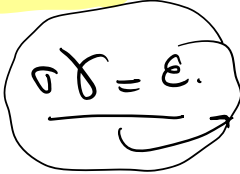
$$\sigma = (1 \ 4 \ 5 \ 2 \ 3) \text{ is a 5-cycle}$$

$$2. \ \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 4 & 7 & 1 & 6 & 5 & 2 & 3 \end{pmatrix} \in S_7$$

$$\tau = (1 \ 4 \ 6 \ 2 \ 7 \ 3) \text{ is a 6-cycle}$$

How do we find the inverse of a cycle? **Easy! Just reverting its action:**

THEOREM 2.13. If σ is an r -cycle, then σ^{-1} is also an r -cycle. Moreover, if $\sigma = (k_1 k_2 \dots k_{r-1} k_r)$, then $\sigma^{-1} = (k_r k_{r-1} \dots k_2 k_1)$.

Proof. Let $\gamma = (k_r k_{r-1} \dots k_2 k_1)$. We need to show that $\gamma\sigma = \text{id}$ and $\sigma\gamma = \text{id}$. (here)  see the course notes.

Goal: $(\gamma\sigma)(k) = k$ for all $k \in \mathbb{X}_n$.

We consider two cases:

CASE 1 $k \notin \{k_1, k_2, \dots, k_r\}$

In this case k is fixed by both σ and γ . Thus:

$$(\gamma\sigma)(k) = \gamma(\sigma(k)) = \gamma(k) = k$$

CASE 2, $k \in \{k_1, \dots, k_r\}$

• If $k = k_i$ for $i = 1, \dots, r-1$

$$\begin{aligned} \underline{(\gamma\sigma)(k_i)} &= \gamma(\sigma(k_i)) = \gamma(k_{i+1}) = \\ &= \underline{k_i} \end{aligned}$$

•) For $k = k_r$ we have that

$$\begin{aligned} \underline{(\gamma\sigma)(k)} &= \gamma(\sigma(k_r)) = \gamma(k_1) = \\ &\quad \uparrow \\ &\quad k = k_r \end{aligned}$$

$$= k_r = k$$



How do we find the product of two cycles?

1. We can write them using the matrix notation, find their product as before, and then express the result using cycle notation.
2. We can use cycle notation.

For example, take $\alpha = (2 \ 4 \ 5)$ and $\beta = (1 \ 2 \ 4)$ in S_5 .

Using matrix notation we obtain

$$\begin{aligned}\alpha\beta &= (2 \ 4 \ 5)(1 \ 2 \ 4) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 3 & 5 & 2 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 1 & 5 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 3 & 1 & 2 \end{pmatrix} = (1 \ 4)(2 \ 5) \quad \text{next video}\end{aligned}$$

Using cycle notation we have that

$$\overbrace{(2 \ 4 \ 5)}^{\alpha} \overbrace{(1 \ 2 \ 4)}^{\beta} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 3 & 1 & 2 \end{pmatrix}$$